

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CSC 3116 – Computer Networks (Lab)

Lab 1: IP Addressing I

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1. Learning Objectives

By the end of this lab, you should be able to:

1. Distinguish between physical (MAC), logical (IP), and port addresses.
2. Represent IPv4 addresses in binary and dotted-decimal notation.
3. Convert 8-bit binary numbers to decimal and vice versa.
4. Determine validity and ranges of IPv4 addresses.

2. Address Types in a Network

When a process on one host communicates with a process on another host, three levels of addresses are involved: physical, logical, and port addresses.

2.1 Physical (MAC) Address

- Hardware address of a network interface, defined by the LAN technology (e.g., Ethernet, Wi-Fi).
- Ethernet uses a 48-bit (6-byte) address, usually written as 12 hexadecimal digits, e.g., CA-18-85-0C-B7-2C.
- First 24 bits: Organizationally Unique Identifier (OUI); last 24 bits: device-specific ID.
- Burned into the NIC ROM; used inside a LAN for frame delivery.

2.2 Logical (IPv4) Address

- IPv4 uses a 32-bit logical address for each host or router interface.
- Written as four 8-bit octets separated by dots (dotted-decimal), e.g.,

10000000 00001011 00000011 00011111 $\Rightarrow$ 128.11.3.31.

- Address space size: $2^{32} = 4,294,967,296$ unique addresses.

2.3 Port Address

- 16-bit number identifying a process on a host (e.g., a web server or mail server).
- Managed by IANA for well-known services, for example:

Port	Service	Protocol
20, 21	File Transfer Protocol (FTP)	TCP
25	Simple Mail Transfer Protocol (SMTP)	TCP
80	Hypertext Transfer Protocol (HTTP)	TCP
110	Post Office Protocol v3 (POP3)	TCP
443	Secure HTTP (HTTPS)	TCP

3. Binary ↔ Decimal Conversion (8 bits)

Each IPv4 octet is an 8-bit unsigned integer in the range $[0, 255]$.

3.1 Positional Weights

For an 8-bit number $b_7b_6b_5b_4b_3b_2b_1b_0$, the positional weights are:

Bit position	7	6	5	4	3	2	1	0
Weight	128	64	32	16	8	4	2	1

3.2 Binary to Decimal

Algorithm: initialize $\text{sum} = 0$; for each bit b_i , if $b_i = 1$ add 2^i to sum .

Example 1. Convert 10011001_2 to decimal.

$$10011001_2 = 1 \cdot 128 + 0 \cdot 64 + 0 \cdot 32 + 1 \cdot 16 + 1 \cdot 8 + 0 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 153.$$

Example 2. Convert 01001101_2 to decimal.

$$01001101_2 = 0 \cdot 128 + 1 \cdot 64 + 0 \cdot 32 + 0 \cdot 16 + 1 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 77.$$

3.3 Decimal to Binary (0–255)

Algorithm using weights $\{128, 64, 32, 16, 8, 4, 2, 1\}$:

- 1) Let d be the decimal number and w the current weight.
- 2) If $d \geq w$, write bit 1 and set $d := d - w$; else write bit 0.
- 3) Move to the next smaller weight and repeat.

Example 3. Convert 172_{10} to 8-bit binary.

$$172 \Rightarrow 10101100_2.$$

Example 4. Convert 20_{10} to 8-bit binary.

$$20 \Rightarrow 00010100_2.$$

4. IPv4 Address Notation and Validity

An IPv4 address is written as

$$a.b.c.d$$

where each octet a, b, c, d is an integer in $[0, 255]$.

4.1 Validity Rules

- Exactly four octets separated by dots.
- Each octet is a decimal integer between 0 and 255.
- Avoid ambiguous leading zeros (e.g., “045”); typical notation uses plain decimal.

Example. Check validity of the following addresses:

a) 111.56.045.78

b) 221.34.7.8.20

5. Ranges of Address

Example

Find the number of addresses in a range if the first address is 146.102.29.0 and the last address is 146.102.32.255.

Solution

$$\text{Number of addresses} = (0 \times 256^3 + 0 \times 256^2 + 3 \times 256^1 + 255 \times 256^0) + 1 = 1024$$

Example

The first address in a range of addresses is 14.11.45.96. If the number of addresses in the range is 32, what is the last address?

Solution

$$\text{Last address} = (14.11.45.96 + 0.0.0.31)_{256} = 14.11.45.127$$

6. Lab Exercises

Show all working in the space provided or on an attached sheet.

6.1 Binary ↔ Decimal

1. Convert the following 8-bit binaries to decimal:

- a) 11001010
- b) 11100001
- c) 10101111

2. Convert the following decimals to 8-bit binary:

- a) 21
- b) 50
- c) 65
- d) 169

6.2 IPv4 Address Conversion

1. Convert the following IPv4 addresses from binary to dotted-decimal:

- a) 10000001 00001011 00001011 11101111
- b) 11000001 10000011 00011011 11111111

2. Convert the following IPv4 addresses from dotted-decimal to binary:

- a) 111.56.45.78
- b) 221.34.7.82

10. Review Questions (IP Addressing–I)

Q1. Classification of address

List and briefly define the three types of addresses mentioned in the lecture: *physical address*, *logical address*, and *port address*.

Q2. Physical vs. logical address

Explain why a physical (MAC) address alone is not suitable for identifying a host across an internetwork composed of multiple different LAN technologies.

Q3. MAC address properties

- a) What is the length, in bytes and bits, of an Ethernet MAC address?
- b) How are the Organizationally Unique Identifier (OUI) and Device ID arranged within this address?

Q4. Port address usage

- a) What is the length (in bits) of a port address?
- b) Why is the Internet Assigned Number Authority (IANA) important in the context of port numbers?
- c) Give two examples of well-known ports and their associated applications (for example, HTTP, FTP, SMTP).

Q5. IPv4 address structure

Describe how a 32-bit IPv4 address is organized into octets and written in dotted-decimal form. Use the example

10000000 00001011 00000011 00011111

from the lecture to illustrate your answer.

Q6. Binary to decimal (single octet)

Using the positional weight method from the lecture, convert the following 8-bit binary numbers to decimal:

- a) 11001010
- b) 11100001
- c) 10101111

Q7. Decimal to binary (single octet)

Using the method shown in the slides, convert these decimal numbers to 8-bit binary:

- a) 21
- b) 50
- c) 65
- d) 169

Q8. IPv4 dotted-decimal ↔ binary

- a) Convert 192.168.10.25 to binary notation, showing all four octets.
- b) Convert the binary address
11110000.10101010.11100011.00011100
to dotted-decimal notation.

Q9. Validity of IPv4 addresses

For each of the following, state whether it is a valid IPv4 address according to the lecture and justify your answer:

- a) 111.56.045.78
- b) 221.34.7.8.20

Q10. Ranges of address

- a) Using the base-256 method from the slides, compute the number of addresses in the range whose first address is 146.102.29.0 and last address is 146.102.32.255.
- b) Using the same style, find the last address when the first address is 14.11.45.96 and the number of addresses in the range is 32 (that is, add 0.0.0.31 in base 256).

*Adapted from: Official Cert Guide CCNA 200-301, Vol. 1, W. Odom, Cisco Press, 2019;
CCNA Routing and Switching, T. Lammle, Wiley, 2016; and course materials (CSC 3116).*